

QUESTION BANK

Chapter: Rest and Motion

Physics | Multiple Choice Questions (MCQ)

Total Questions: 100 | Each Question Carries Equal Marks

Basic Concepts of Rest & Motion (Q1–Q10)

Q1. An object is said to be at rest when?

- (A) it moves with constant velocity
- (B) it does not change its position with time
- (C) it moves in a straight line
- (D) it has zero acceleration

Q2. An object is in motion when?

- (A) it remains stationary
- (B) its position changes with respect to a reference point over time
- (C) it has constant mass
- (D) it is acted on by no force

Q3. Rest and motion are?

- (A) absolute concepts
- (B) relative concepts
- (C) independent of observer
- (D) not related to reference frame

Q4. Which of the following is an example of rest??

- (A) A rolling ball
- (B) A flying bird
- (C) A book on a table
- (D) A flowing river

Q5. The reference point used to describe motion is called?

- (A) Origin
- (B) Frame of reference
- (C) Coordinate axis
- (D) Displacement

Q6. A passenger sitting in a moving train is at rest with respect to?

- (A) the ground
- (B) the trees outside
- (C) another passenger in the same train
- (D) a person standing on the platform

Q7. Which of the following is NOT a type of motion??

- (A) Translatory
- (B) Rotatory
- (C) Oscillatory

(D) Stationary

Q8. Motion in a straight line is called?

- (A) Circular motion
- (B) Random motion
- (C) Rectilinear motion
- (D) Oscillatory motion

Q9. The motion of the Earth around the Sun is an example of?

- (A) Rectilinear motion
- (B) Circular motion
- (C) Oscillatory motion
- (D) Random motion

Q10. The motion of a pendulum is an example of?

- (A) Translatory motion
- (B) Circular motion
- (C) Oscillatory motion
- (D) Rotatory motion

Distance & Displacement (Q11–Q20)

Q11. Distance is a?

- (A) vector quantity
- (B) scalar quantity
- (C) neither scalar nor vector
- (D) dimensionless quantity

Q12. Displacement is a?

- (A) scalar quantity
- (B) vector quantity
- (C) dimensionless quantity
- (D) unitless quantity

Q13. The SI unit of distance and displacement is?

- (A) cm
- (B) km
- (C) m
- (D) mm

Q14. Which can be zero even if distance is non-zero??

- (A) Speed
- (B) Displacement
- (C) Time
- (D) Mass

Q15. A person walks 4 m East and then 3 m North. The displacement is?

- (A) 7 m

- (B) 1 m
- (C) 5 m
- (D) 12 m

Q16. If a body returns to its starting point, the displacement is?

- (A) equal to total distance
- (B) double the distance
- (C) zero
- (D) negative

Q17. Distance travelled is always?

- (A) equal to displacement
- (B) less than displacement
- (C) greater than or equal to displacement
- (D) independent of displacement

Q18. The shortest path between two points is the?

- (A) distance
- (B) displacement
- (C) speed
- (D) velocity

Q19. A car travels 60 km East and 60 km West. Total distance is?

- (A) 0 km
- (B) 60 km
- (C) 120 km
- (D) 30 km

Q20. The displacement of a body moving in a circular path after one complete revolution is?

- (A) equal to circumference
- (B) equal to diameter
- (C) zero
- (D) equal to radius

Speed & Velocity (Q21–Q30)

Q21. Speed is defined as?

- (A) distance/time
- (B) displacement/time
- (C) time/distance
- (D) distance $\times$ time

Q22. Velocity is defined as?

- (A) distance/time
- (B) displacement/time
- (C) speed/time
- (D) distance $\times$ time

Q23. SI unit of speed and velocity is?

- (A) km/h
- (B) m/s
- (C) cm/s
- (D) km/s

Q24. A body moving with uniform speed in a circle has?

- (A) uniform velocity
- (B) variable velocity
- (C) zero acceleration
- (D) constant displacement

Q25. Average speed is calculated as?

- (A) total displacement / total time
- (B) total distance / total time
- (C) final speed / 2
- (D) initial + final speed

Q26. If a car covers 100 m in 10 s, its average speed is?

- (A) 10 m/s
- (B) 1000 m/s
- (C) 0.1 m/s
- (D) 110 m/s

Q27. A body at rest has speed equal to?

- (A) 1 m/s
- (B) infinity
- (C) zero
- (D) depends on mass

Q28. A train covers 360 km in 4 hours. Its speed in m/s is?

- (A) 90 m/s
- (B) 25 m/s
- (C) 40 m/s
- (D) 100 m/s

Q29. Odometer in a vehicle measures?

- (A) velocity
- (B) speed
- (C) distance
- (D) acceleration

Q30. Speedometer measures?

- (A) instantaneous speed
- (B) average speed
- (C) distance
- (D) displacement

Acceleration (Q31–Q40)

Q31. Acceleration is defined as the rate of change of?

- (A) distance
- (B) speed
- (C) velocity
- (D) displacement

Q32. SI unit of acceleration is?

- (A) m/s
- (B) m/s^2
- (C) km/h
- (D) m^2/s

Q33. If velocity increases uniformly, acceleration is?

- (A) zero
- (B) negative
- (C) positive and constant
- (D) variable

Q34. Deceleration means?

- (A) positive acceleration
- (B) zero acceleration
- (C) negative acceleration
- (D) constant velocity

Q35. A car starts from rest and reaches 20 m/s in 5 s. Acceleration is?

- (A) 100 m/s^2
- (B) 4 m/s^2
- (C) 25 m/s^2
- (D) 0.25 m/s^2

Q36. A body moving with uniform velocity has acceleration equal to?

- (A) 1 m/s^2
- (B) depends on speed
- (C) zero
- (D) 10 m/s^2

Q37. If a body decelerates from 30 m/s to 0 in 10 s, retardation is?

- (A) 300 m/s^2
- (B) 3 m/s^2
- (C) 0.3 m/s^2
- (D) 30 m/s^2

Q38. Negative acceleration is called?

- (A) velocity
- (B) deceleration / retardation
- (C) displacement

(D) jerk

Q39. For uniform circular motion, the magnitude of acceleration?

- (A) is zero
- (B) changes direction but not magnitude
- (C) both magnitude and direction change
- (D) neither changes

Q40. A freely falling body has acceleration equal to (approx.)?

- (A) 10 m/s^2
- (B) 9.8 cm/s^2
- (C) 9.8 m/s^2
- (D) 980 m/s^2

Equations of Motion (Q41–Q50)

Q41. First equation of motion is?

- (A) $v = u + at$
- (B) $s = ut + \frac{1}{2}at^2$
- (C) $v^2 = u^2 + 2as$
- (D) $s = vt$

Q42. Second equation of motion is?

- (A) $v = u + at$
- (B) $s = ut + \frac{1}{2}at^2$
- (C) $v^2 = u^2 + 2as$
- (D) $a = (v-u)/t$

Q43. Third equation of motion is?

- (A) $v = u + at$
- (B) $s = ut + \frac{1}{2}at^2$
- (C) $v^2 = u^2 + 2as$
- (D) $s = (u+v)/2 \times t$

Q44. A body starts from rest ($u = 0$) with $a = 5 \text{ m/s}^2$. Velocity after 4 s is?

- (A) 5 m/s
- (B) 20 m/s
- (C) 40 m/s
- (D) 2 m/s

Q45. A body with $u = 10 \text{ m/s}$ decelerates at 2 m/s^2 . It stops after?

- (A) 5 s
- (B) 20 s
- (C) 2 s
- (D) 10 s

Q46. Distance covered in 3 s by a body starting from rest with $a = 4 \text{ m/s}^2$ is?

- (A) 18 m

- (B) 36 m
- (C) 6 m
- (D) 12 m

Q47. Using $v^2 = u^2 + 2as$, if $u = 0$, $a = 10 \text{ m/s}^2$, $s = 5 \text{ m}$, then $v = ?$

- (A) 5 m/s
- (B) 100 m/s
- (C) 10 m/s
- (D) 50 m/s

Q48. Which equation gives displacement without using time??

- (A) $v = u + at$
- (B) $s = ut + \frac{1}{2}at^2$
- (C) $v^2 = u^2 + 2as$
- (D) $s = vt$

Q49. A car brakes from 20 m/s to rest. If $a = -4 \text{ m/s}^2$, braking distance is?

- (A) 50 m
- (B) 5 m
- (C) 100 m
- (D) 25 m

Q50. If $a = 0$ and $u = 15 \text{ m/s}$, displacement in 3 s is?

- (A) 5 m
- (B) 45 m
- (C) 30 m
- (D) 15 m

Graphical Representation (Q51–Q60)

Q51. Slope of a distance-time graph gives?

- (A) acceleration
- (B) displacement
- (C) speed
- (D) velocity

Q52. Slope of a velocity-time graph gives?

- (A) distance
- (B) displacement
- (C) speed
- (D) acceleration

Q53. Area under a velocity-time graph gives?

- (A) acceleration
- (B) displacement
- (C) speed
- (D) force

Q54. A straight line parallel to time axis on a distance-time graph indicates?

- (A) uniform motion
- (B) rest
- (C) accelerated motion
- (D) deceleration

Q55. A straight line with positive slope on a distance-time graph indicates?

- (A) rest
- (B) uniform motion
- (C) accelerated motion
- (D) deceleration

Q56. A curved distance-time graph indicates?

- (A) uniform motion
- (B) rest
- (C) non-uniform motion
- (D) zero velocity

Q57. A horizontal line on a velocity-time graph means?

- (A) rest
- (B) uniform acceleration
- (C) uniform velocity
- (D) deceleration

Q58. Area of a triangle in a v-t graph represents?

- (A) average velocity
- (B) displacement
- (C) acceleration
- (D) speed

Q59. On a v-t graph, a straight line through origin indicates?

- (A) uniform velocity
- (B) uniform acceleration from rest
- (C) rest
- (D) non-uniform motion

Q60. A velocity-time graph for uniform deceleration is?

- (A) horizontal line
- (B) vertical line
- (C) straight line with positive slope
- (D) straight line with negative slope

Types of Motion (Q61–Q70)

Q61. A ball thrown vertically upward shows?

- (A) uniform motion
- (B) uniform acceleration throughout
- (C) deceleration going up, acceleration going down

(D) constant speed

Q62. Motion along a curved path is called?

- (A) rectilinear motion
- (B) curvilinear motion
- (C) oscillatory motion
- (D) rotatory motion

Q63. Motion of blades of a fan is?

- (A) translatory
- (B) oscillatory
- (C) rotatory
- (D) rectilinear

Q64. Motion of molecules in a gas is?

- (A) rectilinear
- (B) circular
- (C) random
- (D) periodic

Q65. A spinning top exhibits?

- (A) translatory motion only
- (B) rotatory motion
- (C) oscillatory motion
- (D) rectilinear motion

Q66. Uniform motion means?

- (A) constant acceleration
- (B) constant speed in a straight line
- (C) changing direction at constant speed
- (D) zero displacement

Q67. Non-uniform motion means?

- (A) constant speed
- (B) varying speed or direction
- (C) zero acceleration
- (D) rectilinear path only

Q68. Periodic motion repeats itself after a fixed?

- (A) distance
- (B) displacement
- (C) time interval
- (D) speed

Q69. Simple Harmonic Motion (SHM) is a type of?

- (A) random motion
- (B) translatory motion
- (C) periodic oscillatory motion
- (D) rotatory motion

Q70. A stone dropped from a height undergoes?

- (A) uniform motion
- (B) uniform acceleration (free fall)
- (C) deceleration
- (D) zero motion

Relative Motion (Q71–Q75)

Q71. Two trains moving in the same direction at 60 km/h and 40 km/h. Their relative speed is?

- (A) 100 km/h
- (B) 20 km/h
- (C) 60 km/h
- (D) 40 km/h

Q72. Two trains moving in opposite directions at 60 km/h and 40 km/h. Their relative speed is?

- (A) 20 km/h
- (B) 40 km/h
- (C) 60 km/h
- (D) 100 km/h

Q73. A person walking at 4 m/s boards a train moving at 10 m/s in the same direction. His velocity relative to ground is?

- (A) 6 m/s
- (B) 14 m/s
- (C) 4 m/s
- (D) 10 m/s

Q74. The concept of relative motion shows that motion is?

- (A) absolute
- (B) relative
- (C) always uniform
- (D) always accelerated

Q75. A passenger in a bus sees trees moving backward. This is an example of?

- (A) absolute motion
- (B) relative motion
- (C) oscillatory motion
- (D) random motion

Newton's Laws & Momentum (Q76–Q85)

Q76. Newton's First Law is also called the Law of?

- (A) Gravitation
- (B) Inertia
- (C) Acceleration
- (D) Action-Reaction

Q77. A body continues in rest or uniform motion unless acted on by?

- (A) gravity
- (B) friction only
- (C) an external unbalanced force
- (D) mass

Q78. Inertia of a body depends on its?

- (A) velocity
- (B) acceleration
- (C) mass
- (D) shape

Q79. Newton's Second Law relates force to?

- (A) velocity
- (B) distance
- (C) rate of change of momentum
- (D) inertia

Q80. Unit of force in SI system is?

- (A) Dyne
- (B) Newton
- (C) Joule
- (D) $\text{kg}\cdot\text{m}$

Q81. If net force on a body is zero, the body?

- (A) must be at rest
- (B) must be moving
- (C) is in equilibrium (at rest or uniform motion)
- (D) accelerates

Q82. Momentum is defined as?

- (A) mass $\times$ acceleration
- (B) mass $\times$ velocity
- (C) force $\times$ time
- (D) force $\times$ distance

Q83. SI unit of momentum is?

- (A) $\text{N}\cdot\text{s}$ or $\text{kg}\cdot\text{m}/\text{s}$
- (B) N/s
- (C) $\text{kg}\cdot\text{m}/\text{s}^2$
- (D) J

Q84. Newton's Third Law states that action and reaction are?

- (A) equal in magnitude, same direction
- (B) equal in magnitude, opposite direction
- (C) unequal and opposite
- (D) equal and act on same body

Q85. A rocket propels itself using?

- (A) Newton's First Law
- (B) Newton's Second Law
- (C) Newton's Third Law
- (D) Law of Gravitation

Practical & Miscellaneous (Q86–Q100)

Q86. A car travelling at 72 km/h is equivalent to how many m/s??

- (A) 72 m/s
- (B) 20 m/s
- (C) 7.2 m/s
- (D) 200 m/s

Q87. Convert 10 m/s to km/h:?

- (A) 10 km/h
- (B) 100 km/h
- (C) 36 km/h
- (D) 3.6 km/h

Q88. A body has zero velocity but non-zero acceleration. This is possible when?

- (A) it moves uniformly
- (B) it is at the highest point of vertical throw
- (C) it moves in a circle
- (D) it decelerates to rest and stays

Q89. Which instrument measures the distance travelled by a vehicle??

- (A) Speedometer
- (B) Odometer
- (C) Barometer
- (D) Accelerometer

Q90. The rate of change of displacement is?

- (A) speed
- (B) acceleration
- (C) velocity
- (D) momentum

Q91. If distance-time graph is a parabola, the motion is?

- (A) uniform
- (B) at rest
- (C) uniformly accelerated
- (D) random

Q92. A body starts from rest and accelerates uniformly. The distance covered in successive seconds are in ratio?

- (A) 1:2:3:4
- (B) 1:3:5:7
- (C) 1:4:9:16

(D) 2:4:6:8

Q93. The area under a speed-time graph gives?

- (A) acceleration
- (B) distance
- (C) displacement only
- (D) velocity

Q94. A boy runs 4 m North and 3 m East. His distance and displacement are?

- (A) 7 m, 5 m
- (B) 5 m, 7 m
- (C) 7 m, 7 m
- (D) 5 m, 5 m

Q95. Which of the following always has the same direction as velocity??

- (A) Acceleration
- (B) Displacement
- (C) Force
- (D) Momentum

Q96. The velocity of light is approximately?

- (A) 3×10^8 m/s
- (B) 3×10^{10} m/s
- (C) 3×10^{11} m/s
- (D) 3×10^{12} m/s

Q97. Sound travels at approximately ___ m/s in air at room temperature?

- (A) 330 m/s
- (B) 3000 m/s
- (C) 30 m/s
- (D) 3 m/s

Q98. A body covers equal distances in equal time intervals. It has?

- (A) zero velocity
- (B) uniform motion
- (C) non-uniform motion
- (D) zero displacement

Q99. Which of the following is a vector quantity??

- (A) Speed
- (B) Distance
- (C) Mass
- (D) Velocity

Q100. Centripetal acceleration acts?

- (A) along the velocity
- (B) away from centre
- (C) towards the centre of circular path
- (D) perpendicular to the plane

ANSWER KEY

Q1	B	Q2	B	Q3	B	Q4	C
Q5	B	Q6	C	Q7	D	Q8	C
Q9	B	Q10	C	Q11	B	Q12	B
Q13	C	Q14	B	Q15	C	Q16	C
Q17	C	Q18	B	Q19	C	Q20	C
Q21	A	Q22	B	Q23	B	Q24	B
Q25	B	Q26	A	Q27	C	Q28	B
Q29	C	Q30	A	Q31	C	Q32	B
Q33	C	Q34	C	Q35	B	Q36	C
Q37	B	Q38	B	Q39	B	Q40	C
Q41	A	Q42	B	Q43	C	Q44	B
Q45	A	Q46	A	Q47	C	Q48	C
Q49	A	Q50	B	Q51	C	Q52	D
Q53	B	Q54	B	Q55	B	Q56	C
Q57	C	Q58	B	Q59	B	Q60	D
Q61	C	Q62	B	Q63	C	Q64	C
Q65	B	Q66	B	Q67	B	Q68	C
Q69	C	Q70	B	Q71	B	Q72	D
Q73	B	Q74	B	Q75	B	Q76	B
Q77	C	Q78	C	Q79	C	Q80	B
Q81	C	Q82	B	Q83	A	Q84	B
Q85	C	Q86	B	Q87	C	Q88	B
Q89	B	Q90	C	Q91	C	Q92	B
Q93	B	Q94	A	Q95	D	Q96	B
Q97	A	Q98	B	Q99	D	Q100	C

— End of Question Bank —